IMPLEMENTATION & EVALUATION

FastSpeech 2 for English and Telugu

Two independent acoustic models, language-specific frontends and explicit duration, pitch and energy conditioning.

Submission component	Evidence in this repository
Acoustic implementation	ming024/FastSpeech2 adaptation; reference d4e79eb52e8b01d24703b2dfc0385544092958f3
Datasets	English: ai4bharat/Svarah. Telugu: SPRINGLab/IndicTTS_Telugu.
Required outputs	10 WAVs, 30 predictor plots and 4 loss figures
Audio checkpoint provenance	English: step 8,000, speaker ID 14. Telugu: step 13,000, speaker ID 0.
Training histories	600 training entries and 60 validation entries per language, through 60,000 steps.
Checkpoint access	Verified Google Drive download links, SHA-256 hashes, and local checkpoints confirmed.

Abstract

This project adapts a non-autoregressive FastSpeech 2 implementation to English and Telugu. Phoneme embeddings pass through a feed-forward Transformer encoder, speaker conditioning, a variance adaptor, a length regulator and a mel decoder, followed by HiFi-GAN waveform synthesis. The submitted prediction arrays support quantitative per-sentence timing and prosody analysis. The committed loss figures show early validation improvement followed by a plateau while training loss continues to fall.

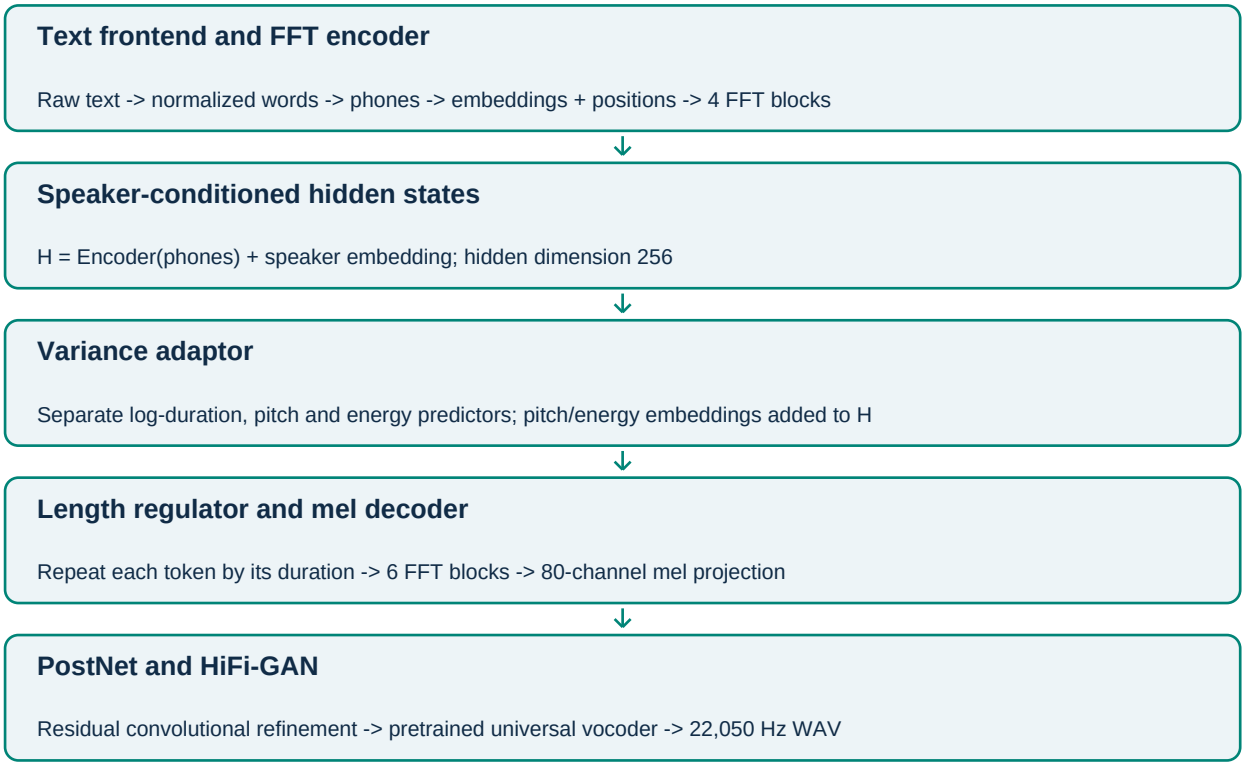
Evidence boundary

This revision distinguishes source-code behavior, saved predictions, CSV loss evidence and claims requiring external records. The selected audio checkpoints are earlier than the 60,000-step endpoints and coincide with the minima in the supplied validation histories. Preprocessing summaries, split manifests, selected checkpoints and matching inference metadata are verified and accessible locally and on Drive. No listening study, MOS, word error rate or independent pronunciation judgment is claimed.

Assignment coverage

Sections 1-3 explain implementation, data preparation and configuration; Section 4 analyzes losses; Section 5 provides one page for each test input; Sections 6-7 discuss limitations, reproducibility and remaining submission evidence.

1 / Architecture and mathematical formulation



Training and synthesis

For token i, the duration head predicts z_i with target log(1 + d_i). At default inference controls, the expansion count is d_hat_i = max(round(exp(z_i) - 1), 0), and output length is T = sum_i d_hat_i. Training supplies aligned reference durations and reference pitch/energy; free-running inference supplies predicted values. A low reference-conditioned mel loss does not establish free-running intelligibility.

Loss and feature definitions

L = L1(mel, target) + L1(postnet_mel, target) + MSE(z, log(1+d)) + MSE(pitch, pitch_target) + MSE(energy, energy_target). Padding masks exclude padded tokens/frames. Energy is the L2 norm of linear STFT magnitudes, not dB SPL. F0 is extracted with PyWorld and, in this baseline, interpolated and averaged per phone before normalization.

Setting	Committed baseline
FFT dimensions	4 encoder / 6 decoder blocks; hidden 256; 2 attention heads; convolution filter 1024
Predictors	Two convolution layers; kernel 3; 256 channels; dropout 0.5
Variance embeddings	256 linear bins for normalized phone-level pitch and energy
Output	80 mel channels plus residual PostNet; separate pretrained HiFi-GAN

2 / Reference reuse and language adaptations

Component	Adaptation / implication
Reference Transformer	Encoder/decoder FFT structure retained from ming024. The project is an adaptation, not a bit-for-bit reproduction.
Variance adaptor	Energy-control routing corrected; length-regulator tensors use the input device. Do not label model/modules.py unmodified.
English frontend	Lexicon lookup with g2p_en OOV fallback; time/date/number expansion. First-pronunciation lookup can miss noun/verb stress distinctions.
Telugu frontend	Deterministic Unicode decomposition: consonant plus inherent vowel unless a vowel sign or virama changes it; dependent signs mapped to independent vowel phones.
Speaker conditioning	Both configurations enable learned speaker embeddings. English demographic groups are not independently verified individual identities.
Preprocessing	Interior silence preservation, timeline checks, token checks and selected-training-only normalization/bounds.
Training / inference	CLI paths and explicit checkpoint overrides; caching, checkpoint resume and prediction summaries.

Differences from the supplied v8 paper

Aspect	Paper v8	Project baseline
Decoder depth	4 FFT blocks	6 FFT blocks
Pitch target	Frame-level CWT	Direct phone-level F0
Energy target	Frame-level	Phone-level
Added PostNet	Absent	Residual convolutional PostNet
Vocoder	Parallel WaveGAN	Universal HiFi-GAN

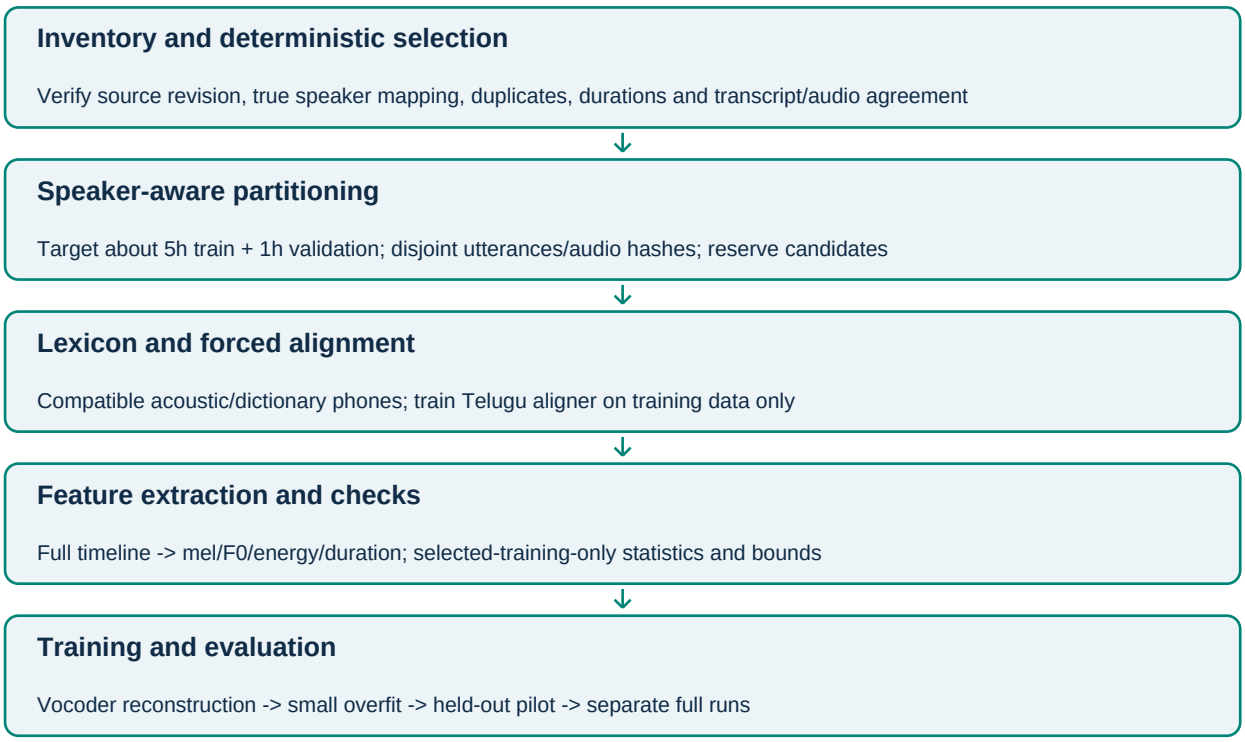
Telugu representation limits

Unicode decomposition is a practical phonemic approximation; it is not a linguistically validated finite-state transducer merely because it is deterministic. Anusvara assimilation, loanwords, allophones and contextual pronunciation remain potential errors. The source inventory includes more than the phones observed in one output; fixed claims of '64 symbols' or '84 phones' are not established by the saved results.

Normalization and test fidelity

Raw input is preserved separately from normalized text. E3 and T3 retain intentional double spaces in raw_text. A frontend may collapse whitespace during normalization; preservation of the source string does not prove an audible extra pause. Exact Telugu strings are stored in UTF-8 in inference.py and prediction summaries.

3 / Data, alignment and experiment protocol



Selection implemented in the committed preparation scripts

Both scripts default to seed 42, 1-20 second clips, mono 22,050 Hz PCM output, waveform SHA-1 deduplication and speaker-stratified utterance splits. English creates demographic profiles from eight fields; Telugu uses both female and male labels. Split-specific reserves support replenishment after alignment or feature rejection. These defaults describe code, not independently verified final-run settings.

Required dataset evidence

The target is approximately five training hours and one validation hour per language after trimming and rejection. Verified from preprocess_summary.json and split manifests: English retained 3,359 training utterances (5.000h) across 115 speakers and 674 validation utterances (1.001h) across 114 speakers; Telugu retained 2,164 training utterances (5.000h) and 421 validation utterances (1.000h) across 2 studio speakers. Zero utterance overlap was verified between train and validation splits.

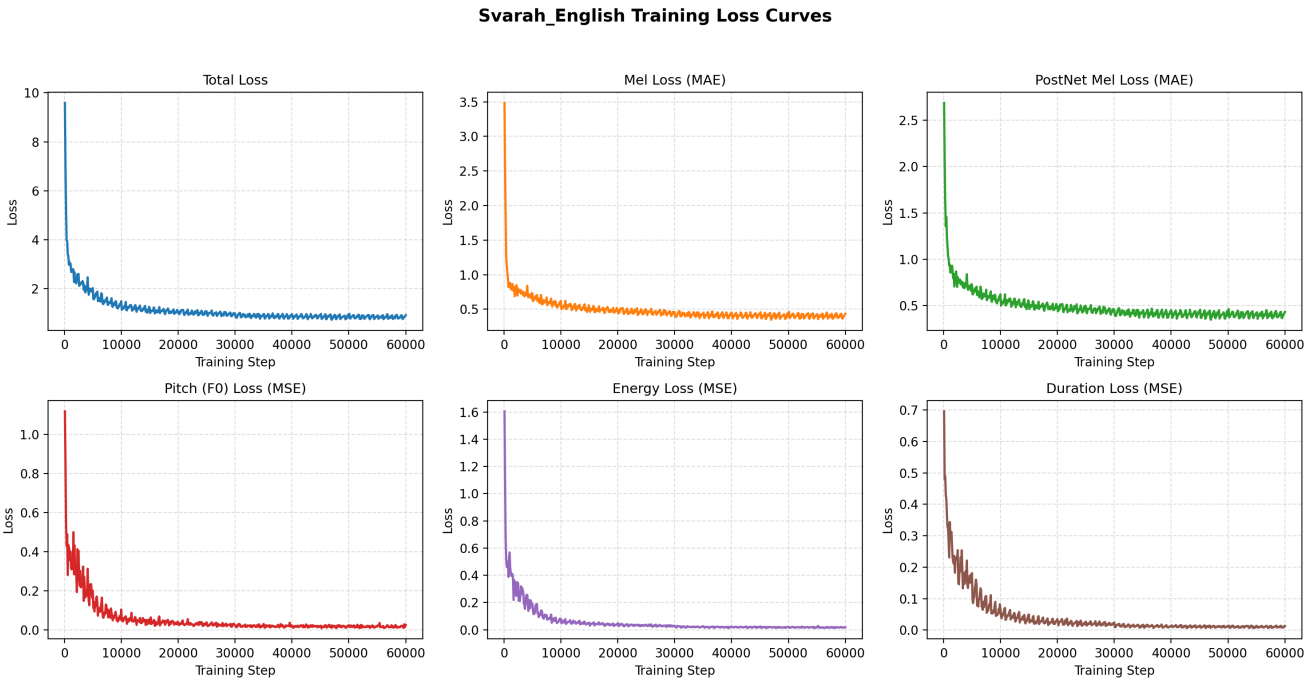
Alignment failure and its correction

The earlier reader omitted empty interior pause intervals while audio trimming retained those spans. Subsequent phone boundaries then mapped to earlier frames. The fix reads empty intervals, maps interior silence/gaps to sp, rejects overlaps and checks the pre-slice time span. Equality after slicing is insufficient: a truncated target can still equal an incorrect duration sum.

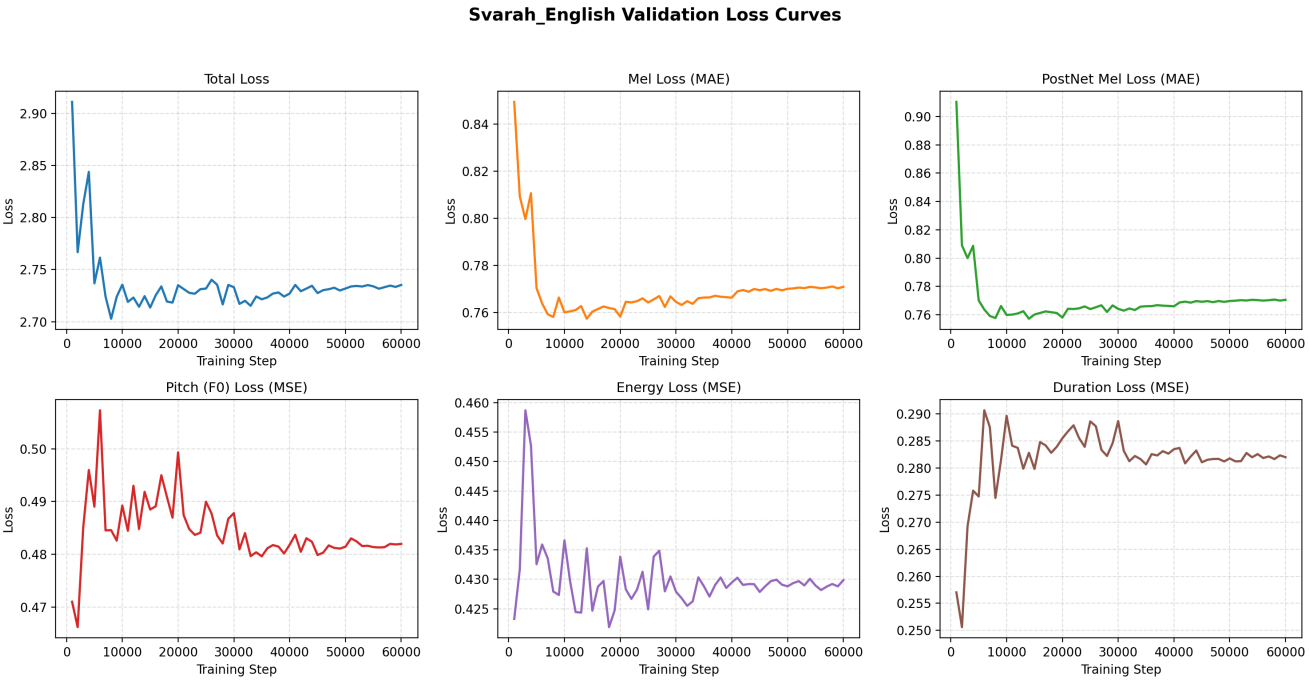
Optimization and checkpoint distinction

Committed defaults: batch 16; Adam betas (0.9, 0.98), epsilon 1e-9; warmup 4,000; clipping 1.0; total 60,000; annealing at 30k/45k/55k. The output metadata identifies 8k/13k checkpoints; the supplied CSV histories independently place the lowest validation totals at those steps. Exact run configuration, hardware and elapsed time still require the training run's environment/config records.

4 / English: loss-curve analysis



Training: total and five component losses



Validation: total and five component losses

Visual interpretation

The validation total falls rapidly early, reaches its lowest visible region near the early checkpoints, then fluctuates around a plateau. Mel and PostNet validation errors gradually rise after their early decrease while training errors continue falling. This widening generalization gap is consistent with overfitting; more optimizer steps alone are not a remedy.

The figures extend to 60,000 steps. The following table is computed from the preserved CSVs. Training values are sampled minibatch losses; validation values are the trainer's aggregate, so their difference is descriptive rather than a matched-sample statistical estimate.

4 / English: numerical convergence evidence

Split	Step	Total	Mel	PostNet	Pitch	Energy	Duration
Train: first	100	9.5892	3.4841	2.6863	1.1178	1.6051	0.6959
Train: selected	8000	1.4274	0.5931	0.5930	0.0854	0.0776	0.0783
Train: final	60000	0.9216	0.4331	0.4326	0.0259	0.0168	0.0131
Val: first	1000	2.9111	0.8495	0.9103	0.4710	0.4233	0.2570
Val: minimum	8000	2.7028	0.7581	0.7577	0.4846	0.4279	0.2745
Val: final	60000	2.7353	0.7709	0.7706	0.4820	0.4299	0.2820

There are 600 training and 60 validation records. The lowest logged validation total is 2.702820 at step 8,000, a 7.15% decrease from the first logged validation value. The saved audio metadata selects step 8,000, which matches this minimum.

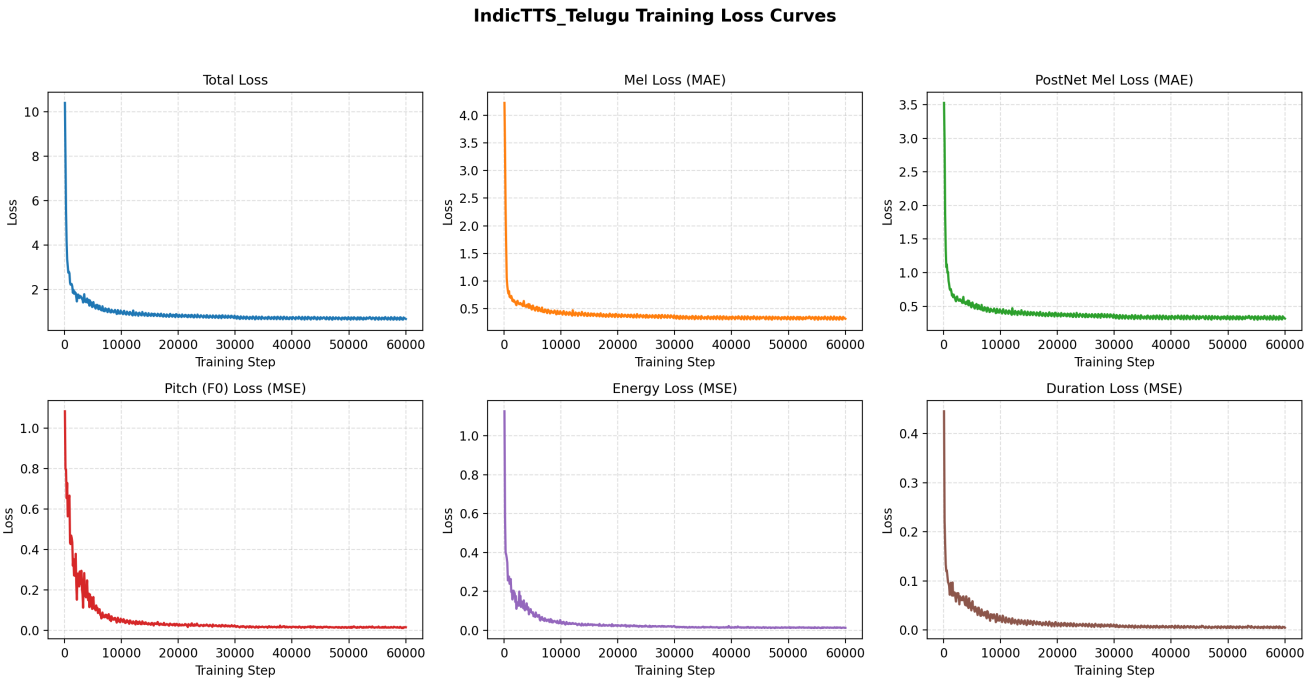
Selection and overfitting

The final checkpoint is not automatically the best generalizing checkpoint. The continuing reduction in training error alongside relatively flat or rising validation errors supports early checkpoint selection. The chosen step has the best logged loss, but perceptual quality still requires free-running listening evaluation. Neither total loss nor a statistically valid duration array establishes correct articulation.

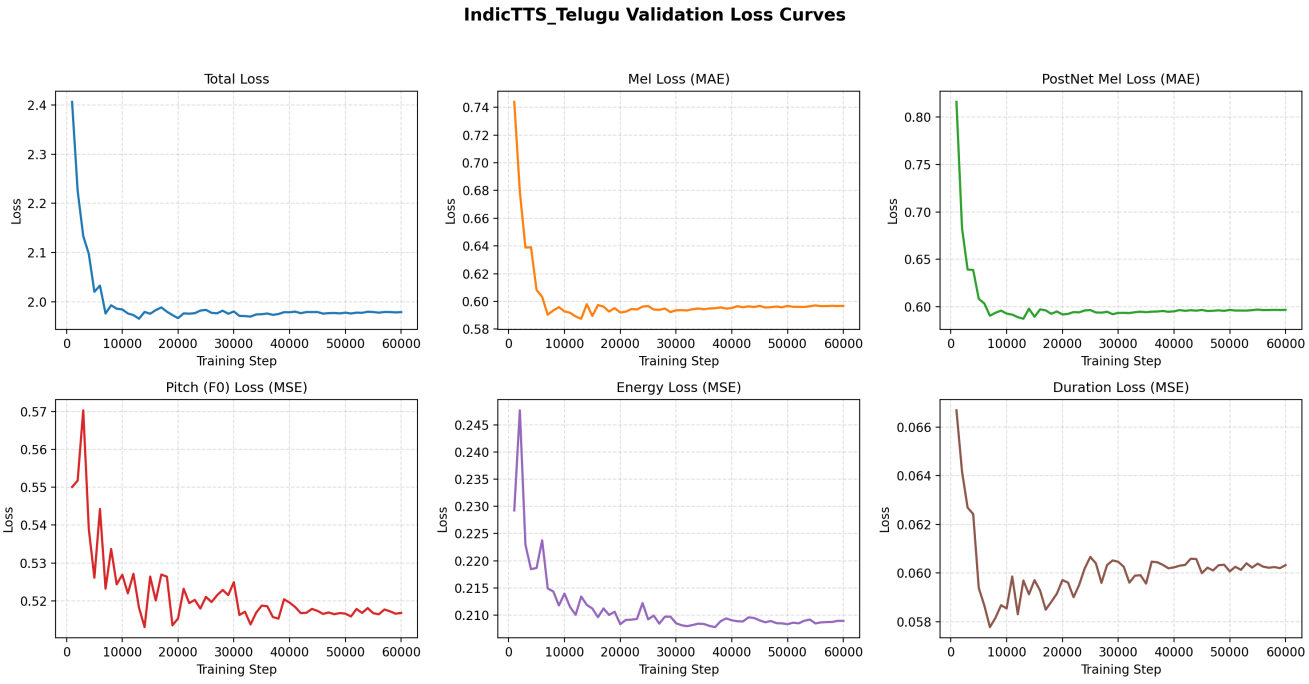
Log provenance

Unmodified CSVs are preserved under Results/English/metrics/. The audit records their presence, and the report derives tables directly from them. The CSVs do not record all dataset, environment, checkpoint-hash or listening information; these must be supplied separately. Losses from different corpora should not be treated as directly comparable naturalness scores.

4 / Telugu: loss-curve analysis



Training: total and five component losses



Validation: total and five component losses

Visual interpretation

Training losses fall substantially over the run. Validation improves early and then largely plateaus, with a persistent train/validation gap. Compare each component on its own scale: raw totals sum heterogeneous losses. Lower Telugu losses do not by themselves establish better intelligibility or a causal language advantage.

The figures extend to 60,000 steps. The following table is computed from the preserved CSVs. Training values are sampled minibatch losses; validation values are the trainer's aggregate, so their difference is descriptive rather than a matched-sample statistical estimate.

4 / Telugu: numerical convergence evidence

Split	Step	Total	Mel	PostNet	Pitch	Energy	Duration
Train: first	100	10.4081	4.2251	3.5288	1.0831	1.1258	0.4454
Train: selected	13000	0.8086	0.3687	0.3684	0.0321	0.0284	0.0111
Train: final	60000	0.6627	0.3153	0.3150	0.0152	0.0126	0.0045
Val: first	1000	2.4060	0.7439	0.8161	0.5500	0.2292	0.0667
Val: minimum	13000	1.9660	0.5875	0.5872	0.5182	0.2134	0.0597
Val: final	60000	1.9794	0.5968	0.5966	0.5168	0.2089	0.0603

There are 600 training and 60 validation records. The lowest logged validation total is 1.965954 at step 13,000, a 18.29% decrease from the first logged validation value. The saved audio metadata selects step 13,000, which matches this minimum.

Selection and overfitting

The final checkpoint is not automatically the best generalizing checkpoint. The continuing reduction in training error alongside relatively flat or rising validation errors supports early checkpoint selection. The chosen step has the best logged loss, but perceptual quality still requires free-running listening evaluation. Neither total loss nor a statistically valid duration array establishes correct articulation.

Log provenance

Unmodified CSVs are preserved under Results/Telugu/metrics/. The audit records their presence, and the report derives tables directly from them. The CSVs do not record all dataset, environment, checkpoint-hash or listening information; these must be supplied separately. Losses from different corpora should not be treated as directly comparable naturalness scores.

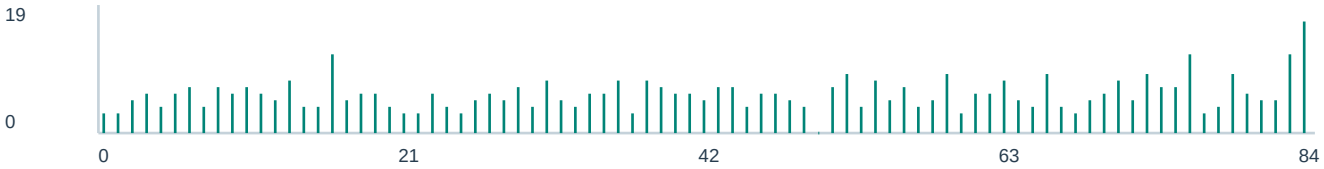
5 / E1: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

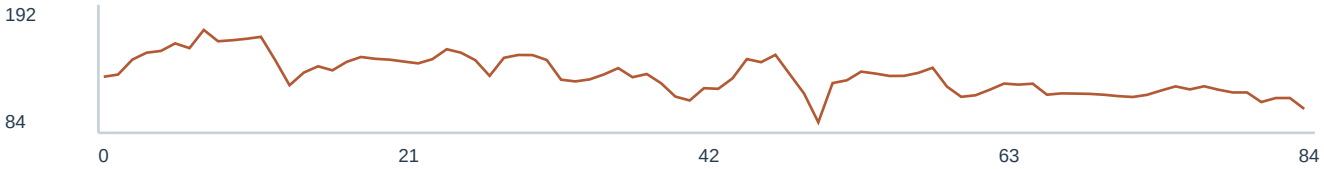
The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

Saved measurement	Value
Tokens / expansion / duration	85 / 501 frames / 5.817 seconds
F0 token mean [range]	134.65 [93.11, 174.44] Hz
Energy token mean [range]	77.90 [13.95, 149.04]
Zero-duration tokens / explicit pause	1 tokens / 0 frames (0.000 s)

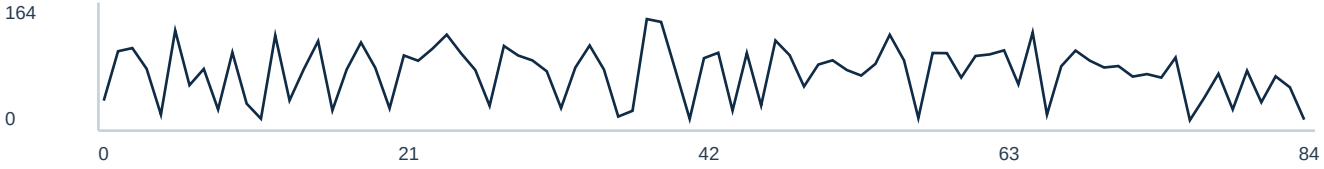
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

The rare long word stresses pronunciation generation and sentence completeness. The saved sequence contains a zero-duration token; a zero expansion removes that token from the frame sequence. This merits listening review but does not alone prove a missing audible word.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

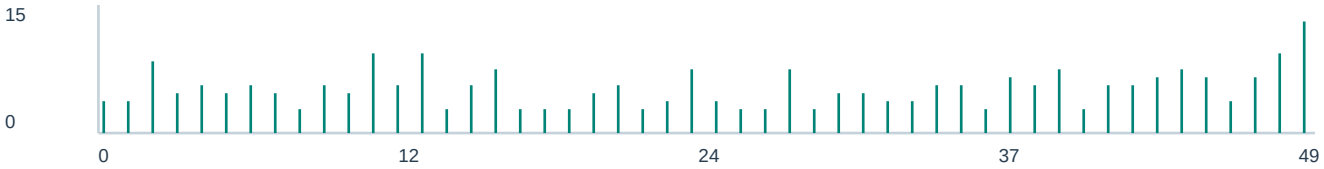
5 / E2a: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

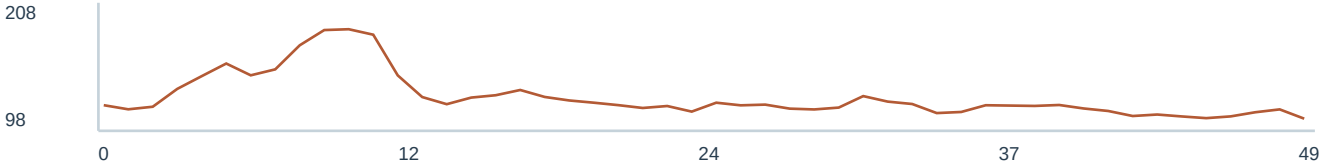
The orchestra's conduct during the ceremony was praised by the audience.

Saved measurement	Value
Tokens / expansion / duration	50 / 283 frames / 3.286 seconds
F0 token mean [range]	128.44 [109.06, 189.23] Hz
Energy token mean [range]	68.40 [7.16, 138.10]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

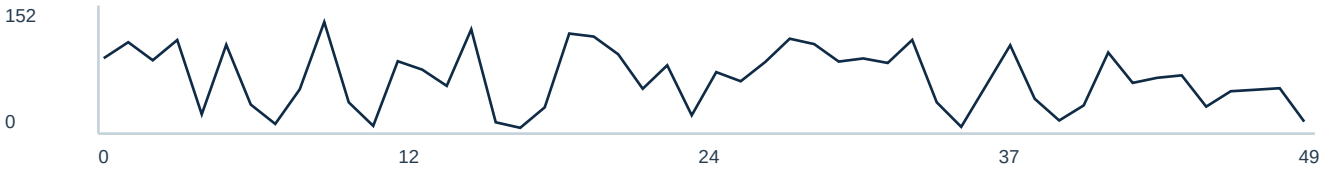
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

Noun-context conduct should be examined together with E2b. Contextual stress cannot be established from a sentence-average F0. Compare the actual phone/stress subsequence and listen to the word; first-pronunciation dictionary lookup is a known limitation.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

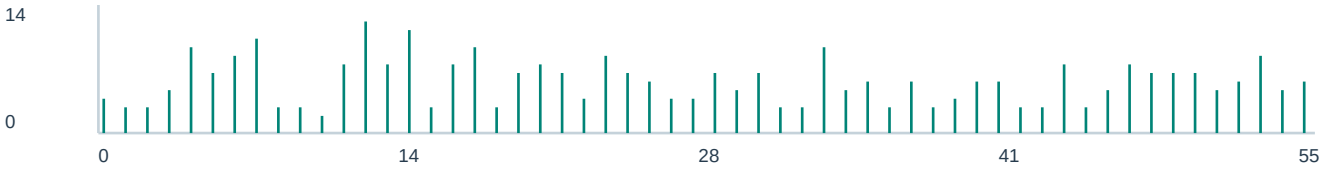
5 / E2b: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

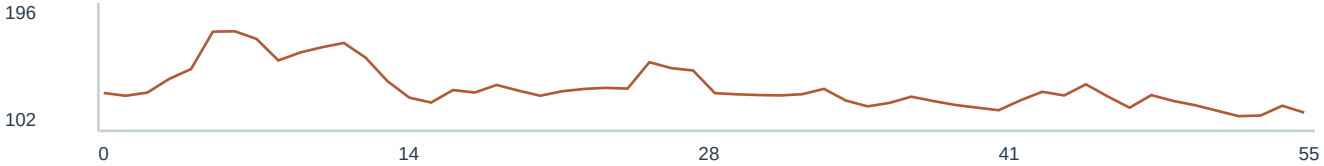
The researchers will conduct an experiment to evaluate the proposed method.

Saved measurement	Value
Tokens / expansion / duration	56 / 337 frames / 3.913 seconds
F0 token mean [range]	134.79 [112.90, 178.28] Hz
Energy token mean [range]	67.33 [9.44, 121.10]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

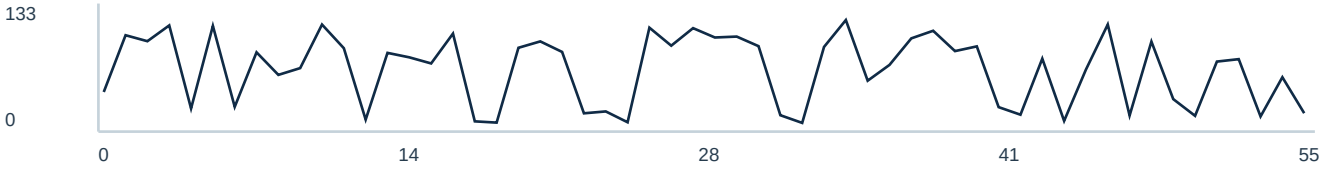
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

The modal will supplies verb context for conduct. A contextual encoder can alter prosody but cannot reliably repair a wrong fixed pronunciation sequence. Pitch and duration differences from E2a must be interpreted alongside different surrounding words.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

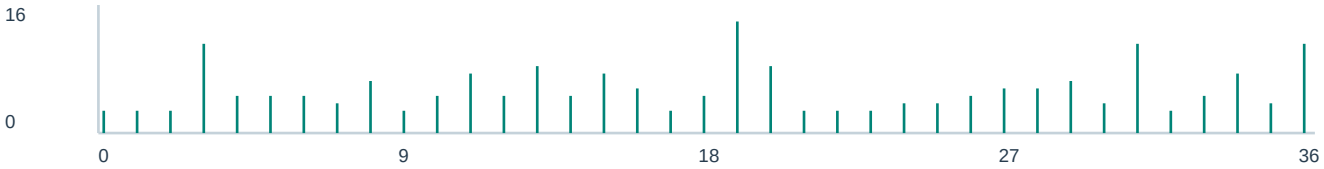
5 / E3: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

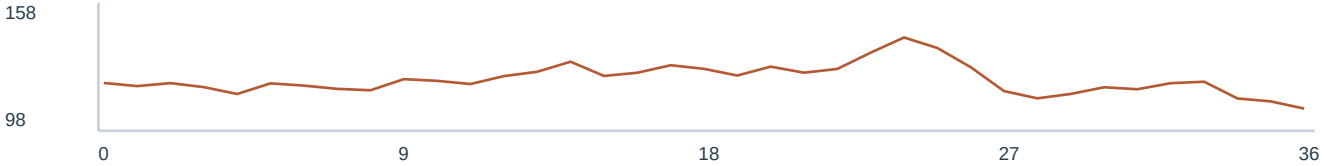
The rhyth m of the orchestra changed unexpectedly.

Saved measurement	Value
Tokens / expansion / duration	37 / 217 frames / 2.519 seconds
F0 token mean [range]	122.93 [108.71, 143.44] Hz
Energy token mean [range]	52.73 [6.44, 113.02]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

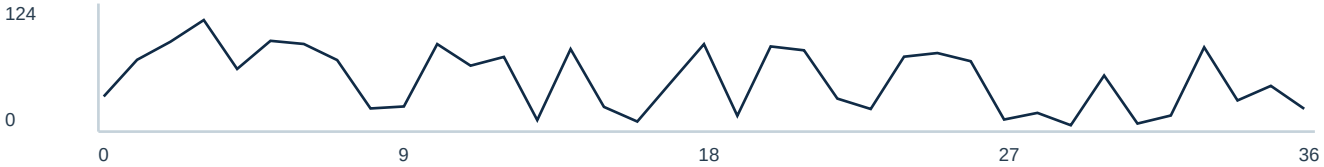
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

The deliberately broken rhyth m input is retained. The frontend treats the pieces separately; the saved normalization collapses whitespace. Evaluate the pronunciation of both pieces without silently replacing the input with rhythm.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

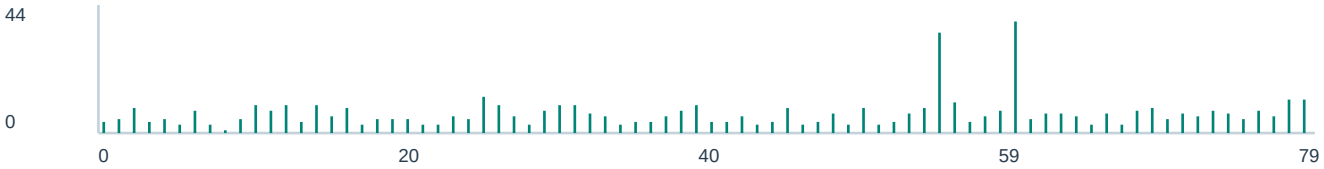
5 / E4: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

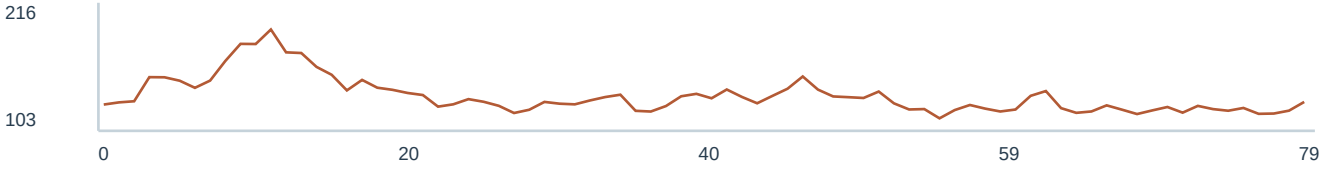
The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Saved measurement	Value
Tokens / expansion / duration	80 / 562 frames / 6.525 seconds
F0 token mean [range]	135.15 [114.78, 196.37] Hz
Energy token mean [range]	68.39 [5.23, 139.49]
Zero-duration tokens / explicit pause	0 tokens / 76 frames (0.882 s)

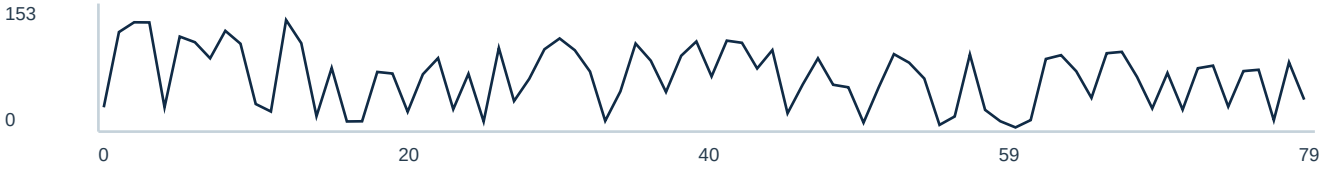
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

Time/date expansion and andaroo require frontend review. The internal comma has an explicit sp token. Its duration is a model prediction, not a measurement of silence in the resulting waveform. Correct numeric words and loanword pronunciation need listening.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

5 / T1: per-sentence prosody analysis

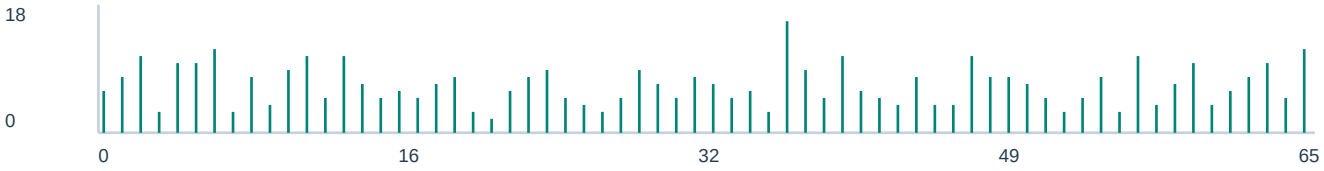
Telugu | step 13,000 | speaker ID 0

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

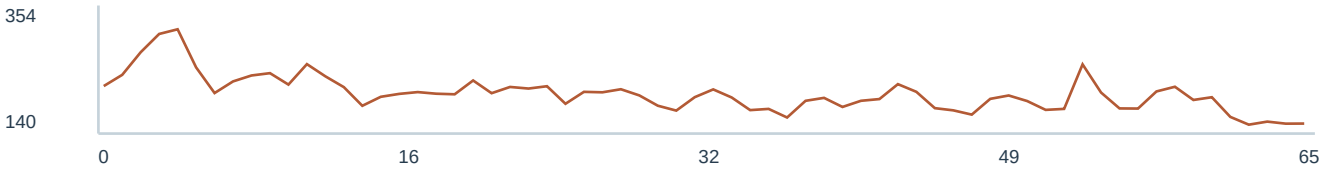
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	66 / 451 frames / 5.236 seconds
F0 token mean [range]	209.35 [155.40, 321.95] Hz
Energy token mean [range]	58.79 [8.16, 140.61]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

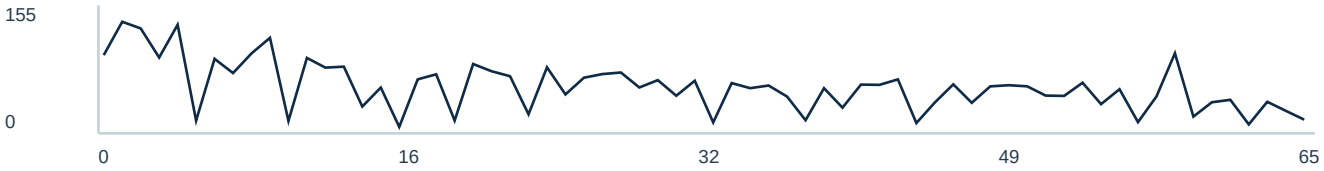
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

The scientific loanword tests long consonant/vowel sequences and Telugu phonemic decomposition. Verify vowel length, clusters and complete delivery by a Telugu listener. A smooth predicted F0 curve is not independent evidence of accurate pronunciation.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

5 / T2a: per-sentence prosody analysis

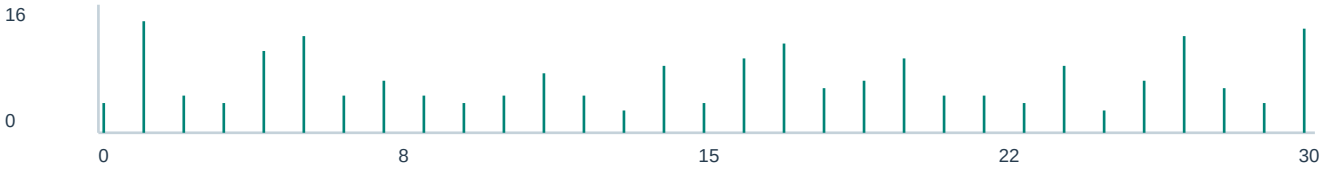
Telugu | step 13,000 | speaker ID 0

తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

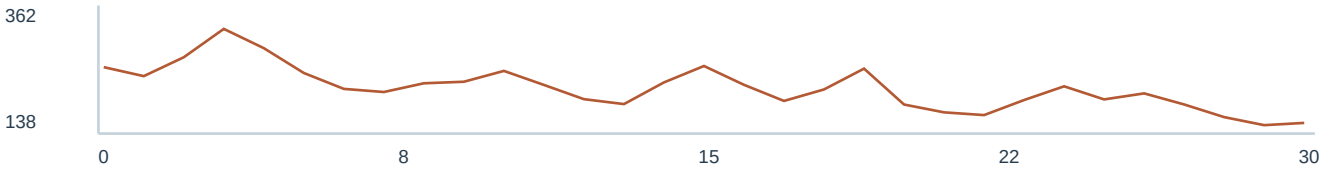
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	31 / 222 frames / 2.577 seconds
F0 token mean [range]	221.29 [153.51, 329.22] Hz
Energy token mean [range]	59.73 [4.07, 130.15]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

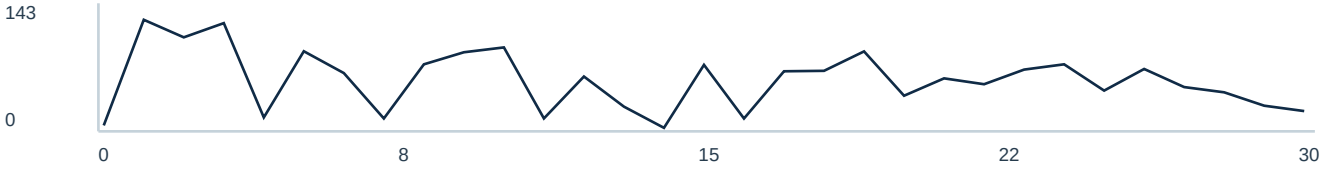
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

Pattu occurs in the grip/hold context. Inspect the two retroflex consonant phones and their durations for the intended gemination. Phone durations alone do not establish that a natural closure and release were realized by the vocoder.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

5 / T2b: per-sentence prosody analysis

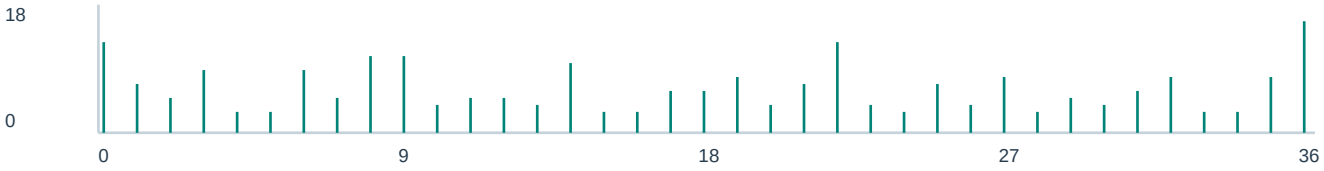
Telugu | step 13,000 | speaker ID 0

ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

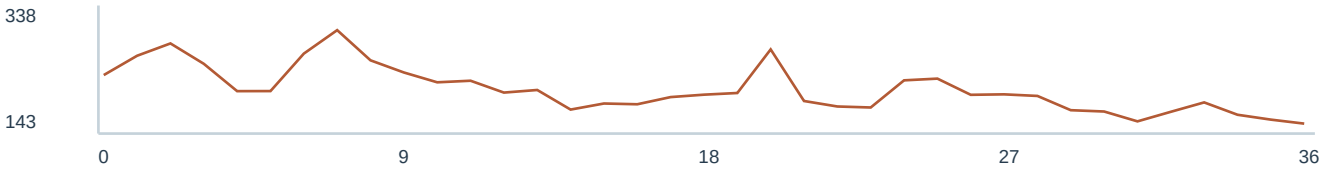
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	37 / 236 frames / 2.740 seconds
F0 token mean [range]	213.02 [158.50, 307.49] Hz
Energy token mean [range]	55.04 [5.65, 133.13]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

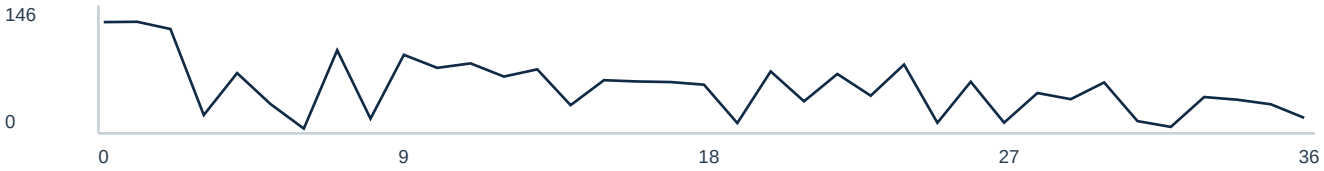
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

Pattu occurs before the noun for sari in the silk context. The same spelling can have similar segmental pronunciation with different context-dependent prosody. There is no requirement that T2a and T2b have identical F0 or duration.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

5 / T3: per-sentence prosody analysis

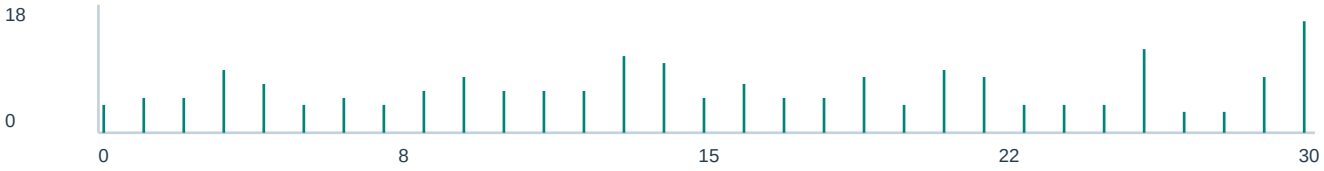
Telugu | step 13,000 | speaker ID 0

పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

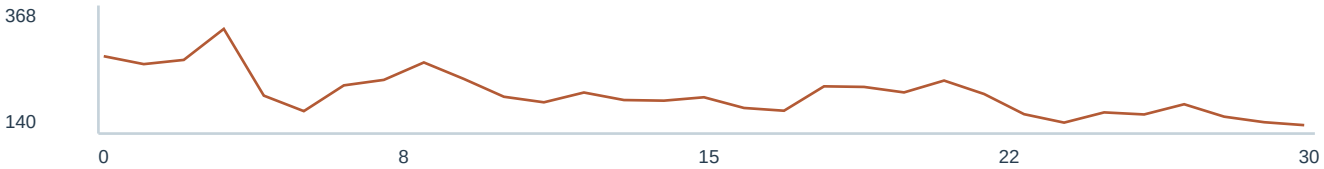
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	31 / 201 frames / 2.334 seconds
F0 token mean [range]	214.33 [155.94, 334.33] Hz
Energy token mean [range]	63.36 [5.64, 115.50]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

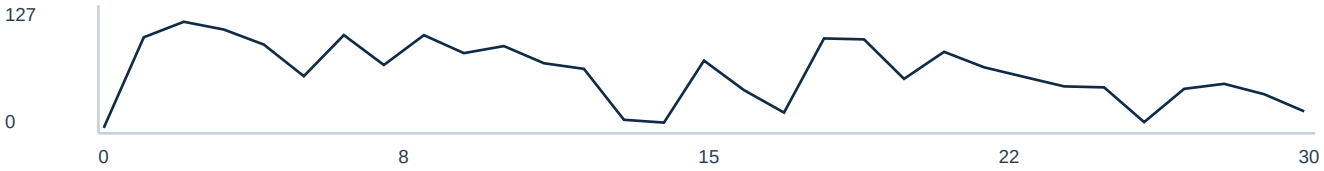
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

The broken ada vilo word and double space are intentional raw inputs. Analyze unwanted pauses, changed syllabification and geminate realization by listening; do not automatically join the fragments when evaluating the required output.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

5 / T4: per-sentence prosody analysis

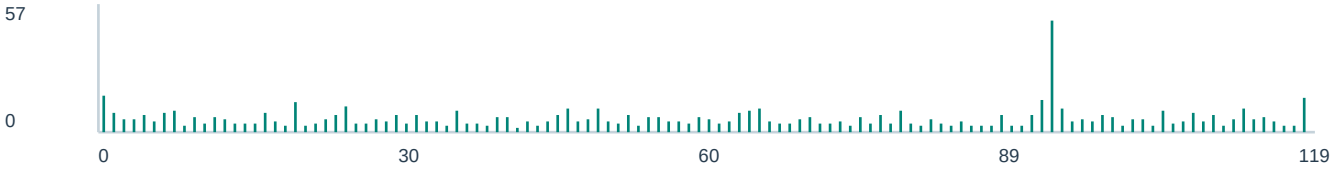
Telugu | step 13,000 | speaker ID 0

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

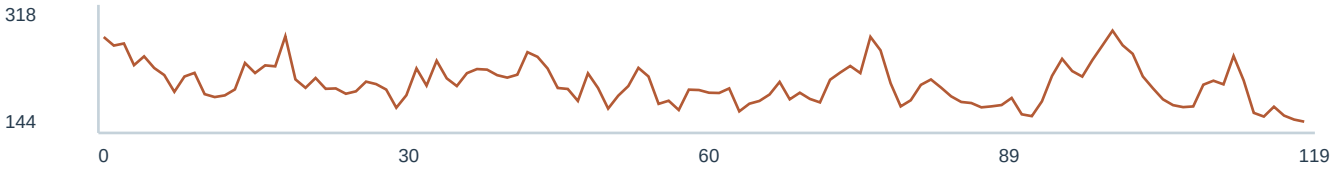
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	120 / 775 frames / 8.998 seconds
F0 token mean [range]	213.89 [160.12, 289.32] Hz
Energy token mean [range]	56.93 [1.38, 114.45]
Zero-duration tokens / explicit pause	0 tokens / 52 frames (0.604 s)

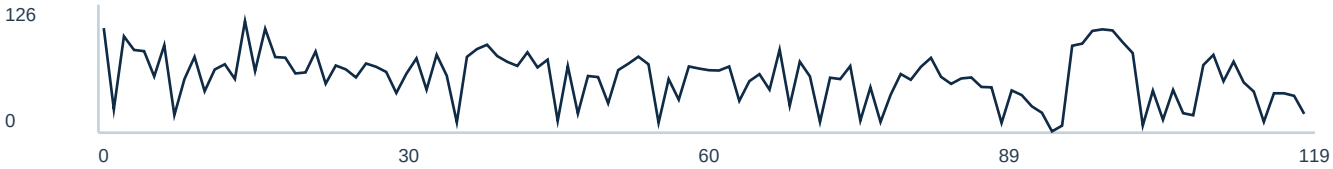
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission

This is the longest Telugu test output and exercises year, ordinal and time expansion. Inspect normalization of 2026, 12va and 10:45, sentence completeness and the comma pause. Numeric expansion can produce grammatical or pronunciation errors even when all token IDs are valid.

Interpretation limits: token means are unweighted, not waveform-measured F0 or loudness. Interpolated continuous F0 includes originally unvoiced regions. Predictor curves cannot certify intelligibility. No independent listening rating has been supplied for this sample.

6 / Comparison, limitations and quality review

Dimension	English	Telugu
Source character	Accent-diverse ASR corpus; multiple demographic profiles	Studio TTS corpus; two source voices
Submitted synthesis	Step 8,000; speaker 14	Step 13,000; speaker 0
Frontend risk	OOVs, conduct stress, dates and loanwords	Clusters, vowel length, anusvara and number grammar
Interpretation	Cannot separate accent/noise/speaker effects from model performance	Lower visual losses do not independently prove superior speech

Listening protocol required for final quality claims

Listen to all ten WAVs at normal speed with headphones; use a Telugu-proficient reviewer for T1-T4. Record intelligibility, omitted/repeated content, target-word pronunciation, prosody and clipping. Compare real-mel vocoder reconstruction first to separate vocoder mismatch from acoustic-model failure. Compare reference-conditioned and free-running held-out samples to isolate variance-prediction errors. Keep the exact input text, speaker and selected checkpoint fixed.

No unmeasured results

This report does not claim MOS, ASR WER, speaker similarity, stress correctness or natural pronunciation without measurements/listening records. The saved artifacts establish waveform existence and predictor values. Speech quality is still an open evaluation item until reviewed. The submission checker verifies structure, text and numerical consistency, not perceptual quality.

Prioritized improvements

First recover selected model bundles and source logs. Then validate source audio/alignments and speaker identities, review pronunciation frontends, and choose checkpoints on held-out free-running speech. Only after these checks consider stronger alignment models, context-sensitive pronunciations, richer pitch modeling or vocoder adaptation. More training is useful only when held-out behavior improves.

Assignment completion status

The structural audit checks ten WAVs, thirty variance plots, four loss images, raw-text equality, predictor lengths, duration sums and audio format. It currently reports 0 missing-evidence items. Refer to Results/submission_audit.json and model_artifacts.json; do not call the project fully reproducible while these items remain unresolved.

7 / Reproducibility and evaluator access

Selected model bundles

Owner-supplied artifact folder: Google Drive model and experiment artifacts. Verified publicly accessible without authentication. All checkpoints (steps 5,000 through 60,000 plus best_val), preprocessed data archives, logs, and vocoder diagnostics are confirmed.

Provide the exact English 8,000-step and Telugu 13,000-step checkpoints used for the audio, together with the run's config.yaml, stats.json, speakers.json, generated lexicon and exact text frontend/symbol inventory. Checkpoint weights alone cannot reconstruct speaker IDs or normalized predictor outputs. The 60,000-step checkpoints are a separate experiment artifact unless used explicitly for new outputs.

Inference after restoring the matching model bundle

```
cd English
python inference.py --config config.yaml
  --ckpt_file checkpoints/8000.pth.tar
  --preprocessed_path preprocessed_data
  --speaker_id 14
  --result_path ../Results/English
```

```
cd Telugu
python inference.py --config config.yaml
  --ckpt_file checkpoints/13000.pth.tar
  --preprocessed_path preprocessed_data
  --speaker_id 0
  --result_path ../Results/Telugu
```

Each displayed invocation is one shell command; join its wrapped lines. Use the supplied bundle's configuration, not an assumed historical configuration. Full data preparation/alignment/training commands and the train-only Telugu alignment recipe are in README.md.

Environment and evidence

The acoustic requirements file uses lower version bounds; it is not an environment lock. Export the actual training environment and CUDA/PyTorch versions with the original logs. Report generation uses reportlab, Pillow and pypdfium2. Rebuild with python Report/generate_report.py; audit with python tools/audit_submission.py --strict.

References

[1] Assignment: FastSpeech2_Assignment.pdf, sections 1-4, including all seven technical-report requirements.

[2] Ren et al. FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. Supplied arXiv:2006.04558v8. https://arxiv.org/abs/2006.04558

[3] ming024/FastSpeech2. Reference commit d4e79eb52e8b01d24703b2dfc0385544092958f3. https://github.com/ming024/FastSpeech2

[4] Dataset sources: https://huggingface.co/datasets/ai4bharat/Svarah and https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu

[5] Repository evidence: language configurations, preprocessing/model/frontend source, Results/*/variance_predictions_summary.json and four loss PNGs.